Machine Learning for European Football Recruitment:

A KNN Approach to Player Similarity

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Abstract—*A difficult part of any sport is the acquisition of a player. This is no different for the world's most popular sport, European football. Teams everywhere in the world have had their fair share of transfers that didn't quite turn out how they wanted. A large reason for this is due to traditional scouting methods being a rather subjective assessment. However, with the growing capabilities of technology and the ability to track detailed performance data, there are new opportunities to be had. This study will be showing those new opportunities, exploring how the in-depth data and machine learning can combine for helping recruitment teams conclude an optimal transfer decision. The performance data used in from the 2017/2018 English Premier League season where player, team, and event-based data are all obtained from Wyscout. In addition, feature-engineered data is included into the statistics, involving passing accuracy, passing frequency, forward passing, smart passing, etc. Given the situation of the sport, to account for players' difference in gametime, event statistics are normalized to a per-90-minute basis (the length of each standard football match,) This study uses these features for a nearest neighbor model, utilizing Euclidean distance, to measure players with similar profiles. As an example, this study will use Kieran Trippier, a former defender of Tottenham Hotspur FC, as a reference to further show our methods. This approach lays a foundation and basis for teams to delve into data-driven research to investigate which players are similar and best for them.*

Index Terms—*Machine Learning, Football Analytics, Player Similarity, Nearest Neighbors, Feature Engineering, Player Recruitment.*

I. INTRODUCTION

Player recruitment poses a difficult challenge in football. For the larger, more popular clubs, this challenge is even more daunting. Spending hundreds of millions on transfers for players that can't quite contribute to the squad is a pressing issue for plenty of clubs at this very moment. Some of the biggest clubs that come to mind are Manchester United, Chelsea, and Tottenham, which all have a gross spend on purchasing players of ~ €965 million, ~ €1.55 billion, and ~ €1.05 billion, respectively, in the past five years. With this expenditure on arriving players, the teams have not quite been able to translate this into winning, totaling five trophies combined in those years, in addition to disappointing league results. It goes without saying that the players brought in are missing expectations and not quite fitting in.

This begs the question of why these clubs can't find the right player to uplift their seasons. Oftentimes, scouting and recruitment are done in a traditional sense, where direct observation and experience in the matter will influence the decision, making for a subjective analysis at times. But as football analytics grow, detailed performance data offers a unique opportunity for these clubs to add a quantitative method to their traditional style. A few seasons after the 2017/2018 Premier League season, Tottenham Hotspur ended up losing their star full-back, Kieran Trippier. This one loss would cause the club issues on their defense for many years, and still does now. Machine learning can offer a fix, where methods such as nearest-neighbor can be used to see which other players closely fit Kieran's profile to fill the shoes he left at Tottenham. By standardizing performance features and measuring Euclidean distance, players can be compared using characteristics tracked and analyzed in their style of play.

This study investigates how nearest neighbors can be utilized for player-to-player similarity using the 2017/2018 Premier League event data from Wyscout. Features utilized are tracked within the data and also engineered from the data, in addition to being standardized on a per 90-minute basis.

II. RELATED WORK

A. Event-Based Football Analytics

Pappalardo et al. introduced a public Wyscout dataset containing detailed spatio-temporal soccer match events, including passes, duels, shots, and spatial coordinates associated with individual match events for professional league matches [1]. This standardized event-based data source enables quantitative player performance analysis across consistent match scenarios. Building on this foundation, PlayeRank adopts multi-dimensional event statistics to implement role-aware player evaluation and objective player ranking [2]. Similar to these studies, our research relies entirely on Wyscout event data. We extract six hand-selected performance features, apply StandardScaler standardization, and construct a standardized feature space to support stable player similarity computation for soccer recruitment analysis.

B. Player Performance and Similarity

VAEP framework proposed context-aware action valuation to quantify the practical match contribution of each on-ball behavior instead of relying on basic statistical counts [3]. Furthermore, the Player Vectors method generates player style embeddings from event sequences and uses distance metrics to identify stylistically similar players [4]. Our study also adopts distance-based similarity analysis through a KNN model with Euclidean distance. However, our

implementation differs by utilizing standardized aggregate per-90 performance features rather than learned embeddings or context-weighted action values, delivering a simpler, more transparent player similarity workflow for practical scouting use cases.

C. Playing Styles and Tactical Considerations

Existing research highlights how player performance patterns are influenced by positional roles and in-game tactical responsibilities [5]. Comprehensive football analytics reviews also emphasize that pure statistical comparison lacks interpretability without sufficient tactical context [6]. These conclusions highlight the core limitation of our current model. In our KNN similarity search, all defenders are grouped into a single comparison group without distinguishing detailed positional sub-roles or differentiated tactical demands, leading to generalized similarity results when benchmarking the full-back reference player Kieran Trippier against other defensive players.

D. Research Gap and Contribution

Previous studies have established reliable event-based data sources, action valuation methods, player style modeling, and tactical analysis frameworks for football analytics. While these studies provide methods for analyzing player performance, our research explores how an interpretable nearest-neighbor model can be applied to a historical football recruitment scenario. Instead of designing new machine learning algorithms, our work contributes a practical, tailored application of classic KNN similarity modeling for targeted football recruitment. We conduct a concrete case study centered on Kieran Trippier, identifying his closest statistical matches and comparing their profiles with existing Tottenham players, Serge Aurier and Kyle Walker-Peters. These comparisons show how our nearest-neighbor model can identify statistical differences between players and provide additional insight for scouting decisions. This study provides an initial baseline for data-driven scouting, while future work will incorporate position-specific feature weighting and tactical feature engineering to explore club-specific player fit evaluation.

III. DATASET AND PREPROCESSING

A. Data Sources

For this study, we utilize Wyscout as the source for our performance data, specifically using games in the 2017/2018 Premier League season. The datasets used involve one specific to player names, player IDs, and positions; another for club information and club IDs; another for match information, starting lineups, and substitutions; and another for events that occur in the games, such as passes, shots, duels, and other match events. In total, we have 380 matches analyzed and 643,150 events in total. Something important to note: the ID's in each respective dataset function as a foreign key to connect it to another dataset, allowing us to put everything together.

B. Player Minutes and Eligibility

Given the fact that players do not play a universal number of minutes, we have specifically tracked the number of minutes a player has throughout a season, with a minimum of 900 minutes needed to be involved in the nearest-neighbor model. To track a player's total minutes, we use our match data to show the lineups for each game. Given the lineups, each player is given 90 minutes, unless substituted at a certain time in the game, at which point we take the minute they were substituted out as their minutes for the game. If a player is not in the lineup and is substituted in, to establish their minutes, we give them the amount of minutes equivalent to 90 minus the minute they were substituted in. We also accounted for red cards, at which a player is sent off and no longer accumulates minutes for that game, effectively the same as being substituted out.

C. Feature Engineering

Given Wyscout's dataset, we have engineered certain features, such as forward passes, backward passes, pass accuracy, etc. It is important to note that Wyscout's event data contains positional coordinates that allow us to track the starting and ending locations of each pass, allowing us to analyze whether a pass went forwards or backwards by checking whether the x-coordinate increased or decreased post-pass. Most features have been standardized to a per-90-minute normalization, where a player's stats are averaged to the amount of the feature per game (90 minutes) represented by the equation below.

Equation (1).

$$\text{Event rate per 90} = (\text{Total eligible events} / \text{Estimated playing minutes}) \times 90$$

The model utilizes the following features: Pass_Attempts_per90; Pass_Accuracy; Forward_Passes_per90; Smart_Passes_per90; Duel_per90; Shot_per90.

IV. MACHINE LEARNING METHOD

A. Feature Standardization

Before implementing the nearest-neighbor model, the selected features are standardized using StandardScaler, largely due to the differences in ranges among the features. By standardizing features with larger values (such as pass accuracy), they become comparable to other features and lessen the impact they have on Euclidean distance in a negative way, allowing us to compare players more accurately.

B. Nearest-Neighbor Retrieval

By implementing Sklearn's nearest-neighbor algorithm, the 6 performance features are utilized, and under the following Euclidean distance formula, the model retrieves six players similar to that of the initial player (one of the six is the initial player)

Equation (2).

$$d(a,b) = \sqrt{[\Sigma_i (z_i(a) - z_i(b))^2]}$$

During this study, Kieran Trippier, defender for Tottenham Hotspur at the time of this season, was chosen as our reference player. The meaning behind this is due to the pivotal role of a player such as Trippier. A player who was not only in the squad for his ability to win back the ball, but also for his ability to attack. Trippier's ability on both sides of the ball is what drove him and Tottenham's success. However, when Trippier left Tottenham a few years later, the club saw an immediate decline; lacking adequate defensive structure and attacking creativity, they saw themselves conceding more and scoring less. The gap Trippier left couldn't be filled; although Tottenham tried their best to replace their star man through the transfer market and current squad, no other player matched Trippier's output. In this case, a player similar to Trippier was what Tottenham needed. Using Trippier as the reference player, the nearest-neighbor model now uses Trippier's six standardized performance features against other defenders in the Premier League. In doing so, the model utilizes the Euclidean distance formula to indicate statistical similarities, where the smaller the distance, the more similar the players are. These insights provide an initial assessment of how the nearest-neighbor model could help augment scouting methods.

V. PRELIMINARY RESULTS AND DISCUSSION

When using Trippier, the model shows:

	shortName	position	minutes_played	Pass_Attempts_per90	Pass_Accuracy	Forward_Passes_per90	Smart_Passes_per90	Duel_per90	Shot_per90
61	K. Trippier	Defender	1915	61.613577	79.938978	30.313316	1.409922	17.718016	0.281984

Five Most Similar Players:

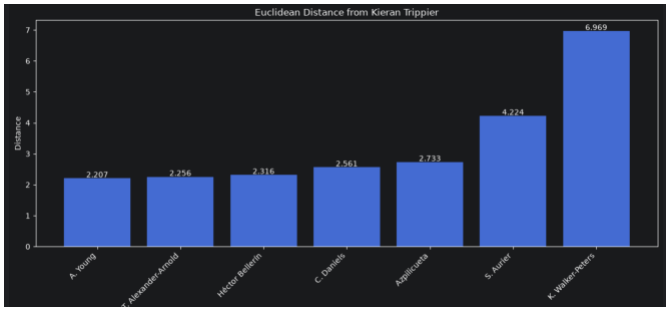
	shortName	position	minutes_played	Pass_Attempts_per90	Pass_Accuracy	Forward_Passes_per90	Smart_Passes_per90	Duel_per90	Sho
28	A. Young	Defender	2447	50.020433	79.117647	30.049040	0.919493	18.978341	
119	T. Alexander-Arnold	Defender	1576	61.332487	77.001862	34.321066	0.913706	15.989848	
108	Héctor Bellerín	Defender	2961	55.562310	84.956236	26.231003	1.155015	17.933131	
67	C. Daniels	Defender	3006	47.335329	78.874130	28.712575	0.898204	14.341317	
86	Azpilicueta	Defender	3330	73.324324	86.951714	48.621622	1.000000	16.432432	

According to the model, Ashley Young, Trent Alexander-Arnold, Hector Bellerin, Charlie Daniels, and Azpilicueta are the closest to Trippier, in that respective order. The closer a player is, the more they statistically match the reference player. Therefore, it is possible one of these five players could have proved to be a potential scouting candidate to fill Trippier's empty shoes.

Alternatively, looking at the replacements Tottenham did go with when Trippier left, we see

	shortName	position	minutes_played	Pass_Attempts_per90	Pass_Accuracy	Forward_Passes_per90	Smart_Passes_per90	Duel_per90	Shot_per90	D
0	S. Aurier	Defender	1429	51.393	79.289	27.334	0.756	25.507	0.945	
1	K. Walker-Peters	Defender	188	49.787	83.654	22.021	0.000	30.638	0.000	

These two players, Serge Aurier and Kyle Walker-Peters, never quite filled Trippier's shoes and truthfully left the club soon after due to performance issues, resulting in a lack of playtime for both players. The distance of Aurier and Walker-Peters from Trippier is around 4.2 and 7.0, respectively. Aurier's distance metric is around 1.5x the distance compared to the alternative players the model listed, and Walker-Peters is over triple the distance of most of those same players. For visualization and understanding purposes, please see the following figure:



Aurier and Walker-Peters are, judging by the data given and the output of the model, not quite similar to Trippier - at least not enough to recreate his work. Thus, giving Tottenham an issue of failing to adequately replace their fullback position. (Something to note, Walker-Peters does not satisfy the 900-minute analysis, rather his induction into the study is more exploratory.)

VI. LIMITATIONS AND FUTURE WORK

This study contains all the 2017/2018 Premier League games but does not note games played outside of this league. Players and teams who participated in the Champions League, Europa League, FA Cup, etc. have that part of their season unaccounted for in this study. For a player who was pivotal for teams in other competitions, this can mean missing out on important metrics. Player-minutes are also approximated at a maximum of 90, although it is more than likely that players have played more than that due to additional time added towards the end of halves, meaning the study may be missing a few extra minutes allotted for themselves.

During preprocessing, certain match events contained null values in the substitution data, which were all omitted from our study. This occurrence was in five of the total 380 games analyzed. Additionally, having a similar player to a reference player does not mean they are a perfect fit/replacement. It is possible a player's stats may be matching, but important characteristics of a player are unaccounted for, such as their leadership. It's with this in mind that we say this study can show how machine learning can augment scouting, not replace it.

Moving forward, to improve the model, certain positions will have certain features weighted more/less in their category when calculating distance between players. For example, a forward player will have attacking features such as shots be more important than other defensive features, such as duels. More engineered features can be accounted for, giving players more statistics to be compared to, in comparison to the base six features the study currently accounts for.

VII. CONCLUSION

This study explores how machine learning can augment the current scouting system in European football by utilizing in-depth match data. Through Wyscout's 2017/2018 datasets, performance for players were measured in six features, where most were normalized to a per-90-minute scale to

account for the amount an individual performs said feature in one game.

Using Kieran Trippier as a reference player, the model evaluated five of his nearest neighbors, showing which other players perform closest to him based on the six features chosen. Furthermore, two other players were shown as statistically different from Trippier, both of which could not quite fill in for Trippier once he left – largely due to the difference in playstyle shown in the data.

While this study and model do not definitively tell clubs if a player is bound to succeed at a club, it represents how machine learning can identify potential candidates, advance current scouting methods, and further evaluate players in comparison to one another.

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